

Present and Absent variables

KD-FK(ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$), CK (ID, DP(7, $\beta_{off} < 0$)); Absent: PK, MK

CP-Rem(IS(1, $\beta_{on} < 0$), DT (5, $\beta_{off} > 0$))Und (ID(4, $\beta_{on} > 0$), DT (5, $\beta_{off} > 0$)), ;Absent: App, Ana, Eva, Cre

LT-Beh (NC, DT, (5, $\beta_{off} > 0$), Cog (ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$)) ;Absent: Con

Note A(True)

- The teacher should avoid asking the students to recall the name of the synthetic indicator which promotes retention over understanding of the concept. [Rem, 1-IS, 5-DT]
- In the lines where the teacher was asking the students to recall examples of indicators, the teacher could have asked the students, ‘How can you determine whether a material is acidic or basic?’ This pushes the children to think beyond simple retention of examples and rather apply their knowledge. [App, 3-ID, 8-DP]
- The teacher instructs the students to passively write down definitions. The teacher can instead ask the students to write down their understanding of the concept of indicators. [Beh, 5-DT]

Note B

- The teacher should ask the students to recall and name the examples of indicators multiple times, this enables better retention due to constant repetition.
- The teacher should avoid asking application based question, without enough context as it will confuse and misdirect the students.
- The teacher should encourage thorough note-taking in the class, as writing down the examples and types of indicators, the students create a concise document for revision of the concepts they learn in class.

Present and Absent variables

KD-FK(ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$)), CK (ID, DP(7, $\beta_{off} < 0$)); Absent: PK, MK
 CP-Und (ID(4, $\beta_{on} > 0$), DT (5, $\beta_{off} > 0$)), ;Absent: Rem, App, Ana, Eva, Cre
 LT-Beh (NC, DT, (5, $\beta_{off} > 0$)), Cog (ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$)) ;Absent: Con

Note A(True)

- When teacher is introducing activation of deoxyribonucleotides, the teacher could encourage a back and forth discussion with students to bring engagement in classroom. [Beh,5,DT]
- Throughout the class, teacher could have given relatable analogies and examples to easily relate and visualise the concept of 'activation of deoxyribonucleotides.' [Und, 4, 5, ID, DT]
- One thing the teacher should continue doing is, when the teacher drew the diagram of conversion of AMP to ATP it helped the children's retention of the information by converting a complex process into an equation.[Cog, 4, 7, ID, DP]

Note B(Foil)

- When teacher is introducing activation of deoxyribonucleotides, the direct, structured way the concept was delivered gives students a clear, uninterrupted line of reasoning to follow.
- The teacher should spend more time to establish the precise definition, terminologies and mechanisms of deoxyribonucleotide activation concept this will help in further retention.
- The teacher shifting from verbal explanation to diagrams without enough verbal framing will confuse the children and their attention might drop.

Present and Absent variables

KD-FK(ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$)), CK (ID, DP(7, $\beta_{off} < 0$)), PK(ID(4, $\beta_{on} > 0$), NC); Absent: MK
 CP-Und (ID(4, $\beta_{on} > 0$), DT (5, $\beta_{off} > 0$)), App(ID(4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$)); Absent: Rem, Ana, Eva, Cre
 LT-Beh (NC, DT, (5, $\beta_{off} > 0$), Cog (ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$)), Con (ID (4, $\beta_{on} > 0$), DP (7, $\beta_{off} < 0$))

Note A(True)

- The teacher should continue asking questions such as, “what do you think will happen,” over question that involve recollection as this encourages application of the learned concept rather than passive retention. [Rem, 2-IS, 6-DT]
- The teacher should continue asking application based questions such as “so if this is decreased, what happens to log value?” which requires drawing inferences to promote higher order thinking. [App, 4-ID, 7-DP]
- The teacher should continue promoting group work in students and proposing problems such as, “if we subtract from this cell the increased value, what do you expect?” for them to collectively solve.[Con, 4-ID, 7-DP]

Note B

- The teacher should avoid asking complex questions such as, “what do you think will happen?” and directly state the concept for better retention.
- The teacher should avoid asking application based questions such as “so if this is decreased, what happens to log value?” without enough context as it will confuse and misdirect the students.
- The teacher should avoid group work and rather work out the problems such as, “if we subtract from this cell the increased value, what do you expect?” in a structured manner to the whole class directly for better retention of the concept.